

CSA-1717 (AI)

PROGRAM 13

AIM

To implement the Minimax algorithm for the Nim game and determine the optimal move for the computer.

ALGORITHM

1. Start the program.
2. Set the initial number of sticks.
3. Allow the player to remove 1, 2, or 3 sticks.
4. Check the current number of sticks.
5. Generate all possible moves for the computer.
6. Apply the Minimax algorithm to evaluate each possible move.
7. Select the move with the best score.
8. Repeat the process until all sticks are removed.
9. Declare the winner based on the last move.
10. Stop the program.

PROGRAM

```
def minimax(sticks, maximizing):
    if sticks == 0:
        return -1 if maximizing else 1
    if maximizing:
        best = -float('inf')
        for move in range(1, min(3, sticks) + 1):
            score = minimax(sticks - move, False)
            best = max(best, score)
        return best
    else:
        best = float('inf')
        for move in range(1, min(3, sticks) + 1):
            score = minimax(sticks - move, True)
```

```

        best = min(best, score)
    return best
def best_move(sticks):
    best_score = -float('inf')
    move = 1
    for i in range(1, min(3, sticks) + 1):
        score = minimax(sticks - i, False)
        if score > best_score:
            best_score = score
            move = i
    return move
sticks = 10
print("Nim Game")
while sticks > 0:
    print("Sticks remaining:", sticks)
    player = int(input("Enter number of sticks to remove (1-3): "))
    if player < 1 or player > 3 or player > sticks:
        print("Invalid move")
        continue
    sticks -= player
    if sticks == 0:
        print("Player wins!")
        break
    computer = best_move(sticks)
    print("Computer removes:", computer)
    sticks -= computer
    if sticks == 0:
        print("Computer wins!")
        break

```

OUTPUT

```
Nim Game
Sticks remaining: 10
Enter number of sticks to remove (1-3): 2
Computer removes: 1
Sticks remaining: 7
Enter number of sticks to remove (1-3): 1
Computer removes: 2
Sticks remaining: 4
Enter number of sticks to remove (1-3): 1
Computer removes: 3
Computer wins!
```

RESULT

The Minimax algorithm was successfully implemented for the Nim game, and the optimal moves were selected based on the possible game states.